TMP and BitLocker

Overview

TPM stands for Trusted Platform Module and is a hardware-based security chip that protects cryptographic keys and system integrity. This chip is located on the motherboard or as a CPU integration that securely stores encryption keys, passwords, and certificates by performing key generation, encryption, decryption, and signing.

BitLocker will unlock disks when TPM verifies that the system has booted in a trusted state which results in protected drive encryption keys. TPMs are often used with BitLocker to prevent attacks such as firmware tampering, ransomware, and brute-force attempts. Even when an operating system is compromised, the TPM will provide better security than software-based methods because it is separate from the main operating system.

Organization Implementation

Trusted Platform Module and BitLocker should be implemented when handling sensitive data, remote workers, or when it is required for compliance. The technology is suitable for remote teams where sensitive data on connected devices can be protected, even when the devices are offline. When regulations or government standards require data encryption at rest, TPM and BitLocker should be implemented to avoid regulatory and compliance fines.

TPM and BitLocker should also be implemented once the organization is ready to implement them where the organization would have TPM chips on most devices. Organizations should especially prepare to use TPM and BitLocker if they lack hardware-based security.